

RESEARCH ASSISTANT PROFESSOR

University of Tennessee - Oak Ridge Innovation Institute
Center for Transportation Research (CTR), University of Tennessee Knoxville
National Transportation Research Center (NTRC), Oak Ridge National Laboratory
Knoxville, TN

Email: uahmed@utk.edu

Web: www.linkedin.com/in/uahmed16; <https://utorii.com/usman-ahmed/>;
<https://ctr.utk.edu/person/usman-ahmed/>

UNIVERSITY EDUCATION**Ph.D. Civil Engineering**

University of Toronto, Canada.
Adviser: Dr. Matthew J. Roorda.
Thesis Title: An Agent-Based Freight Business and Logistics Decisions Simulation

M.Sc. Transportation Systems

Technical University of Munich, Germany.
Adviser: Dr. Rolf Moeckel
Thesis Title: Development of a Microscopic Activity Sequence Modeler

B.E. Civil Engineering

National University of Sciences and Technology, Pakistan.

ACADEMIC POSITIONS***Research Assistant Professor***

University of Tennessee - Oak Ridge Innovation Institute, University of Tennessee Knoxville, USA
Since January 2025

Postdoctoral Fellow

Department of Civil & Mineral Engineering, University of Toronto, Canada
June 2022 - December 2024. Adviser: Dr. Matthew J. Roorda

Researcher/ Modeler

Travel Modeling Group, University of Toronto, Canada
September 2023 - December 2024. Adviser: Dr. Eric J. Miller

Research Assistant

Professorship of Travel Behavior, Technical University of Munich, Germany
February 2017 - August 2018. Adviser: Dr. Rolf Moeckel

PROFESSIONAL EXPERIENCE**Highway Engineer**

Prime Engineering & Testing Consultants, Pakistan, September 2014 - September 2015

PATENTS

P1. Shoman, Maged; **Ahmed, Usman**. Multimodal Computer Vision System and Method for Real-Time Navigation Assistance for Visually Impaired Pedestrians. *U.S. Provisional Application No. 63/917,447. File Date: 11/14/2025*

ACADEMIC ACHIEVEMENTS AND AWARDS

- *Project of the Year Award (2023)*, Institute of Transportation Engineers, Toronto. Project: E-cargo Pilot Study in Toronto. (Role: Project Manager).
- *University of Toronto - Sustainable Action Award (2023)*. Awarded to Purolator as a Project Partner (Dr. Ahmed participated as a project manager from the University of Toronto)
- *Richard Soberman Graduate Student Fellowship (2022)*, University of Toronto Transportation Research Institute.
- *Canadian National Scholarship (2021-2022)*, Canadian Transportation Research Forum.
- *Professor Gerald Steuart Graduate Scholarship in Transportation Engineering (2021)*, University of Toronto Transportation Research Institute.
- *Dr. Mazen Hassounah Graduate Scholarship in Mass Events Transportation and Crowd Management (2019)*, University of Toronto Transportation Research Institute.
- *Fortran Traffic Systems Graduate Scholarship in Transportation Research (2018)*, University of Toronto Transportation Research Institute.
- *Academic Merit Based Scholarship (2010-2014)*, National University of Sciences and Technology, Islamabad, Pakistan. Scholarships are awarded to students who achieve a GPA of 3.5/4.0 or higher.

TEACHING HISTORY

Instructor, University of Toronto, Toronto, ON, Canada. January - April 2024

Course: Freight Transportation and ITS Applications.

Co-Instructor, University of Toronto, Toronto, ON, Canada.

- Freight Transportation and ITS Applications.
Co-taught with Dr. Carlos Rivera-Gonzalez (*January - April 2023*)
- Infrastructure for Sustainable Cities.
Co-taught with Dr. Tufayel Chowdhury (*January - April 2022*)
- Introduction to Transportation Systems.
Co-taught with Dr. Sanjana Hossain (*September - December 2021*)

RESEARCH PRODUCTS**Peer-reviewed Journal Articles**

- J16 Assmussen, Katherine; **Ahmed, Usman** (2026). How is Online Shopping Reshaping Activity Patterns? Modeling Activity Behavior using Discrete Continuous Models. *Accepted in Transportation Research Part A: Policy and Practice*.
- J15 **Ahmed, Usman**; Hawkins, Jason (2026). Residential Price Modeling using Spatially Validated Machine Learning Methods: A Comparison Across Geographic Contexts. *Transportation Research Record*.
- J14 Ghizzawi, Farah; **Ahmed, Usman**; Roorda, Matthew (2026). Parking Patterns of Last-Mile Delivery Vehicles: A Comparison Between Light-Duty Trucks and Electric Cargo Tricycles in Downtown Toronto. *Transportation Research Interdisciplinary Perspectives*. <https://doi.org/10.1016/j.trip.2026.101941>
- J13 Amaral, Julia; Holguín-Veras, Jose; Roorda, Matthew J.; ... **Ahmed, Usman**; ... (2026). Freight Trip Generation in Metropolitan Areas: An International Perspective. *Transportation Research Part A: Policy and Practice*. Vol. 204. <https://doi.org/10.1016/j.tra.2025.104847>
- J12 **Ahmed, Usman**; Mousavi, Kianoush; Zhang, Shang; Roorda, Matthew (2025). Noise Impacts and Social Justice Analysis of Off-Peak Deliveries in the Greater Toronto Area. *Transport Policy*. Vol. 167, Pages 66-77. <https://doi.org/10.1016/j.tranpol.2025.03.018>
- J11 Saeedi, Milad; Xu, Junshi; **Ahmed, Usman**; Roorda, Matthew; Hatzopoulou, Marianne (2025). Urban Air Pollution Data Collection, Mapping, and Prediction using Mobile Sensors Installed on Courier Trucks. *Transportation Research Record*. <https://doi.org/10.1177/036119812513242>
- J10 Rivera-Gonzalez, Carlos; **Ahmed, Usman**; Roorda, Matthew (2025). The Evolving Landscape of Urban Logistics: A study of the Greater Golden Horseshoe. *Journal of Transport Geography*, 122, 104055. <https://doi.org/10.1016/j.jtrangeo.2024.104055>
- J9 **Ahmed, Usman**, Moeckel, Rolf (2023). Impact of Life Events on Incremental Travel Behavior Change. *Transportation Research Record*, 2677(9), 594-605. <https://doi.org/10.1177/03611981231159863>
- J8 **Ahmed, Usman**; Hawkins, Jason; Roorda, Matthew (2022). Establishment Location Choice Model Considering Intra-firm Interactions. *Journal of Transport Geography*, 102, 103391. <https://doi.org/10.1016/j.jtrangeo.2022.103391>
- J7 **Ahmed, Usman**; Tu, Ran; Xu, Junshi; Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew (2022). GPS Based Traffic Condition Classification using Machine Learning Approaches. *Transportation Research Record*, 2677(2), 1445-1454. <https://doi.org/10.1177/0361198122111370>
- J6 Hawkins, Jason; **Ahmed, Usman**; Habib, Khandker M. Nurul; Roorda, Matthew (2022). Measuring the Process of Urban Gentrification: A Composite Measurement of the

Gentrification Process in Toronto. *Cities*, 126, 103708. <https://doi.org/10.1016/j.cities.2022.103708>

- J5 **Ahmed, Usman**; Roorda, Matthew (2022). Joint and Sequential Models for Freight Vehicle Type and Shipment Size Choice. *Transportation*, 50(5), 1613-1629. <https://doi.org/10.1007/s11116-022-10289-6>
- J4 **Ahmed, Usman**; Roorda, Matthew (2022). Modelling Carrier Type and Vehicle Type Choice of Small and Medium Size Firms. *Transportation Research Part E: Logistics and Transportation Review*, 160, 102655. <https://doi.org/10.1016/j.tre.2022.102655>
- J3 **Ahmed, Usman**; Roorda, Matthew (2022). Modeling freight Vehicle Type Choice Using Machine Learning and Discrete Choice Methods. *Transportation Research Record*, 2676(2), 541-552. <https://doi.org/10.1177/03611981211044462>
- J2 Xu, Junshi; Tu, Ran; **Ahmed, Usman**; Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew (2021) An Eco-Score System Incorporating Driving Behavior, Vehicle Characteristics, and Traffic Conditions. *Transportation Research Part D: Transport and Environment*, 95, 102866. <https://doi.org/10.1016/j.trd.2021.102866>
- J1 **Ahmed, Usman**; Moreno, Ana; Moeckel, Rolf (2021). Microscopic Activity Sequence Generation: A Multiple Correspondence Analysis to Explain Travel Behavior Based on Socio-Demographic Person Attributes. *Transportation*, 48(3), 1481-1502. <https://doi.org/10.1007/s11116-020-10103-1>

Technical Reports

- TR3 **Ahmed, Usman**; Le, Tho; Rivera-Gonzalez, Carlos; Roorda, Matthew (2024). A Commercial Travel Survey and Its Implications for the Supply Chain in Peel Region and the GTHA.
- TR2 **Ahmed, Usman**; Xu, Junshi; Tu, Ran; Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew (2020). Developing a Scoring System for Eco-Driving: Report II Development of Method for Phase 3.
- TR1 Xu, Junshi; Tu, Ran; **Ahmed, Usman**; Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew (2020). Developing a Scoring System for Eco-Driving: Report I. Literature Review, Development of Method for Phases 1 and 2.

Peer-Reviewed Conference Papers

- C24 **Ahmed, Usman**; Rivera-Gonzalez, Carlos; Roorda, Matthew. Warehouse Location Choice Model Considering Intra-Firm Interactions: A Case Study of the Greater Toronto Area. Presented at the 105th Annual Meeting of Transportation Research Board, January 11-15, 2026, Washington D.C.
- C23 Calderon, Oriana; **Ahmed, Usman**. Joint Modeling of Freight Mode Choice and Shipment Size Using Multi-Output Machine Learning. Presented at the 105th Annual Meeting of Transportation Research Board, January 11-15, 2026, Washington D.C.

- C22 Assmussen, Katherine; **Ahmed, Usman**. How is Online Shopping Reshaping Activity Patterns? Modeling of Activity Behavior using Discrete Continuous Models. *Presented at the 105th Annual Meeting of Transportation Research Board, January 11-15, 2026, Washington D.C.*
- C21 Shoman, Maged; **Ahmed, Usman**. VISIONASSIST: A Comprehensive Framework for Assistive Navigation for Visually Impaired Pedestrians. *Presented at the 105th Annual Meeting of Transportation Research Board, January 11-15, 2026, Washington D.C.*
- C20 Rashidi, Hesam; **Ahmed, Usman**; Nourinejad, Mehdi; Roorda, Matthew. Data-driven Tactical Design in Last-mile Logistics: Considering Driver Preferences in Tour Duration Estimation. *Presented at the 104th Annual Meeting of Transportation Research Board, January 5-9, 2025, Washington D.C.*
- C19 **Ahmed, Usman**; Mousavi, Kianoush; Zhang, Shang; Roorda, Matthew. Noise Impacts of Off-Peak Deliveries in the Greater Toronto Area: Community Noise Survey and Social Justice Analysis. *Presented at the 104th Annual Meeting of Transportation Research Board, January 5-9, 2025, Washington D.C.*
- C18 **Ahmed, Usman**; Hawkins, Jason. Residential Price Modeling with Spatial and Aspatial Cross-Validation: A Comparison Across Geographic Contexts. *Presented at the 104th Annual Meeting of Transportation Research Board, January 5-9, 2025, Washington D.C.*
- C17 Saeedi, Milad; Xu, Junshi; **Ahmed, Usman**; Roorda, Matthew; Hatzopoulou, Marianne. Urban Air Pollution Data Collection, Mapping, and Prediction using Mobile Sensors Installed on Courier Trucks. *Presented at the 104th Annual Meeting of Transportation Research Board, January 5-9, 2025, Washington D.C.*
- C16 Amaral, Julia; Holguin-Veras, Jose; Roorda, Matthew; ... **Ahmed, Usman**; ... Freight Trip Generation in Metropolitan Areas: An International Perspective. *Presented at the 104th Annual Meeting of Transportation Research Board, January 5-9, 2025, Washington D.C.*
- C15 **Ahmed, Usman**; Le, Tho; and Roorda, Matthew. An Urban Freight Model Considering Intra-firm and Supply Chain Relationships. *Presented at the 103rd Annual Meeting of Transportation Research Board, January 7-11, 2024.*
- C14 **Ahmed, Usman**, Xu, Junshi; Ghizzawi, Farah, Saeedi, Milad; Guseva, Lisa; Hatzopoulou, Marianne; Roorda, Matthew. E-cargo Tricycles for Urban Goods Movement: A Comprehensive Analysis of a Pilot Study in Toronto. *Presented at the 103rd Annual Meeting of Transportation Research Board, January 7-11, 2024, Washington D.C.*
- C13 Rivera-Gonzalez, Carlos; **Ahmed, Usman**; Roorda, Matthew. The Evolving Landscape of Urban Logistics: A Study of the Greater Golden Horseshoe. *Presented at the 103rd Annual Meeting of Transportation Research Board, January 7-11, 2024, Washington D.C.*
- C12 Ghizzawi, Farah; **Ahmed, Usman**; Roorda, Matthew. Parking Patterns of Last-Mile Delivery Vehicles: A Comparison Between Light-Duty Trucks and Electric Cargo Tricycles.

Presented at the 103rd Annual Meeting of Transportation Research Board, January 7-11, 2024, Washington D.C.

- C11 **Ahmed, Usman;** Moeckel, Rolf. Analysis of Travel Behavior Change Triggered by Life Events. *Presented at the 16th World Conference on Transport Research, July 17-21, 2023, Montreal, Canada*
- C10 **Ahmed, Usman,** Nourinejad, Mehdi; Roorda, Matthew. Freight Tour Modeling: A Novel Optimization Based Approach. *Presented at the 16th World Conference on Transport Research, July 17-21, 2023, Montreal, Canada*
- C9 **Ahmed, Usman,** Hawkins, Jason; Roorda, Matthew. Firm Location Choice Model Considering Firm Membership. *Presented at the 102nd Annual Meeting of Transportation Research Board, January 8-12, 2023, Washington D.C.*
- C8 **Ahmed, Usman;** Moeckel, Rolf. Impact of Life Events on Incremental Travel Behavior Change *Presented at the 102nd Annual Meeting of Transportation Research Board, January 8-12, 2023, Washington D.C.*
- C7 **Ahmed, Usman;** Nourinejad, Mehdi; Roorda, Matthew. A Tour-Based Model using Continuous Approximation Method. *Presented at the 101st Annual Meeting of Transportation Research Board, January 9-13, 2022, Washington D.C.*
- C6 **Ahmed, Usman;** Tu, Ran, Xu, Junshi; Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew. GPS Based Traffic Condition Classification using Machine Learning Approaches. *Presented at the 101st Annual Meeting of Transportation Research Board, January 9-13, 2022, Washington D.C.*
- C5 **Ahmed, Usman,** Roorda, Matthew. Joint and Sequential Models for Freight Vehicle Type and Shipment Size Choice. *Presented at the 101st Annual Meeting of Transportation Research Board, January 9-13, 2022, Washington D.C.*
- C4 **Ahmed, Usman;** Hawkins, Jason; Roorda, Matthew. Establishment Location Choice Model Considering Intrafirm Interactions. *Presented at the 68th Annual North American Meetings of the Regional Science Association International NARSC – RSAI November 8-13, 2021, Denver, Colorado (Presented Virtually)*
- C3 **Ahmed, Usman;** Roorda, Matthew. Modelling Carrier Type and Vehicle Type Choice of Small and Medium Size Firms. *Presented at the 100th Annual Meeting of Transportation Research Board (Virtual), January 2-29, 2021, Washington D.C.*
- C2 Xu, Junshi; Tu, Ran; **Ahmed, Usman,** Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew. A Scoring System to Evaluate Eco-Driving Using Driving Behaviour and Vehicle Characteristics for Usage-based Insurance. *Presented at the 100th Annual Meeting of Transportation Research Board (Virtual), January 2-29, 2021, Washington D.C.*
- C1 Hawkins, Jason, **Ahmed, Usman,** Habib, Khandker M. Nurul; Roorda, Matthew. Measuring the Process of Urban Gentrification: A Composite Measurement of the

Gentrification Process in Toronto. *Presented at the 66th Annual North American Meetings of the Regional Science Association International NARSC – RSAI November 13-16, 2019, Pittsburgh, Pennsylvania.*

Peer-Review Conference Abstracts

- A21 **Ahmed, Usman**; Miller, Eric. Analysis of dependent-serving activity/travel behavior. *Presented at the Modeling Mobility Conference, Sept 14-17, 2025, Minneapolis, Minnesota.*
- A20 Assmussen, Katherine; **Ahmed, Usman**. How Online Shopping is Reshaping Activity Patterns and Travel Behavior? *Presented at the Modeling Mobility Conference, Sept 14-17, 2025, Minneapolis, Minnesota.*
- A19 **Ahmed, Usman**; Ghizzawi, Farah; Roorda, Matthew 2025. Microhubs in the City of Toronto for Last-mile Deliveries using Electric Cargo Tricycles. *Presented at the International Urban Freight Conference (I-NUF) April 9-11, Los Angeles, California.*
- A18 **Ahmed, Usman**; Roorda, Matthew 2025. An Urban Freight Model with Supply Chain Relationships and Intrafirm Interactions. *Presented at the International Urban Freight Conference(I-NUF) April 9-11, Los Angeles, California.*
- A17 **Ahmed, Usman**; Le, Tho; Roorda, Matthew 2025. A Commercial Travel Survey for the Greater Toronto Area. *Presented at the 13th International Conference on Transport Survey Methods, March 30th - April 4th, 2025, Da Nang, Vietnam.*
- A16 **Ahmed, Usman**; Hassani, Mohamed Saoudi; Tomeo, Francois; Miller, Eric 2024. School Location Choice Modelling for the Montreal Metropolitan Area. *Presented at the 59th Annual Conference of Canadian Transportation Research Forum, May 12 - 15, 2024, Kelowna, British Columbia.*
- A15 **Ahmed, Usman** 2024. Supplier Type Choice Model Considering Distribution Channels. *Presented at the TRB Freight Systems and Marine Young Members Council (YMC-FM) Freight and Marine “Work-in-Progress” Lightning Talks and Poster Session, January 7 - 11, 2024.*
- A14 **Ahmed, Usman**; Hawkins, Jason 2023. Validity of Cross-Validation Methods in Spatial Settings: Case Study of Housing Prices in Rawalpindi and Islamabad Metropolitan Area, Pakistan. *Presented at the 70th Annual North American Meetings of the Regional Science Association International NARSC – RSAI, November 15-18, 2023, San Diego, California.*
- A13 **Ahmed, Usman**; Mousavi, Kianoush; Zhang, Shang; Roorda, Matthew 2023. Social Justice Analysis of Off-Peak Deliveries: A Study of the Greater Toronto Area. *Presented at the 5th VREF Conference on Urban Freight 18-20 October 2023, Gothenburg, Sweden*
- A12 Rivera-Gonzalez, Carlos; **Ahmed, Usman**; Roorda, Matthew 2023. The Evolving Landscape of Urban Logistics: A Study of the Greater Toronto and Hamilton Area.

Accepted for presentation (but withdrawn) at the 5th VREF Conference on Urban Freight 18-20 October 2023, Gothenburg, Sweden

- A11 **Ahmed, Usman**; Xu, Junshi; Ghizzawi, Farah; Saeedi, Milad, Hatzopoulou, Marianne; Roorda, Matthew 2023. E-cargo Tricycle Pilot Study: A Comparison of E-cargo Tricycles and Trucks for Urban Freight Deliveries. *Presented at the NECTAR Cluster 3 (Logistics and Freight) Workshop, 20-22 September 2023, Madrid, Spain.*
- A10 Moreno, Ana; Noulis, Georgina; **Ahmed, Usman**; Schiffer, Maximilian; Moeckel, Rolf 2023. Understanding the Impact of Life Events on Travel Behavior Change via Machine Learning. *Presented at the 25th Euro Working Group on Transportation Meeting 6 - 8 September 2023, Santander, Spain.*
- A9 **Ahmed, Usman**; Le, Tho; Roorda, Matthew 2023. Commercial Travel Survey: Insights of Urban Goods Movement in the GTHA. *Presented at the 58th Annual Conference of Canadian Transportation Research Forum May 7 - 10, 2023, Toronto, Ontario.*
- A8 Farah Ghizzawi; **Ahmed, Usman**; Roorda, Matthew 2023. A Comparison of Parking Behaviours of Delivery Trucks and E-cargo Tricycles. *Presented at the 58th Annual Conference of Canadian Transportation Research Forum May 7 - 10, 2023, Toronto, Ontario.*
- A7 **Ahmed, Usman** 2023. An Electric Cargo Tricycle Pilot Study 2023. *Presented at the TRB Freight Systems and Marine Young Members Council (YMC-FM) Freight and Marine "Work-in-Progress" Lightning Talks and Poster Session January 8 - 12, 2023.*
- A6 **Ahmed, Usman**; Roorda, Matthew 2022. Establishment Location Choice Model Considering Firm Membership. *Presented at 57th Annual Conference of Canadian Transportation Research Forum June 12 - 15, 2022, Montreal, Quebec.*
- A5 **Ahmed, Usman**; Roorda, Matthew 2022. Freight Vehicle Type and Shipment Size Modelling using Discrete Choice Methods. *Presented at 57th Annual Conference of Canadian Transportation Research Forum, June 12 - 15, 2022, Montreal, Quebec*
- A4 **Ahmed, Usman**; Roorda, Matthew 2022. Modelling Freight Vehicle Type and Shipment Size Choice. *Presented at the 9th METRANS International Urban Freight Conference May 25-27, Long Beach, California.*
- A3 **Ahmed, Usman**; Roorda, Matthew 2022. Applying Machine Learning to Model Freight Vehicle Type Choice. *Presented at the 9th METRANS International Urban Freight Conference May 25-27, Long Beach, California.*
- A2 **Ahmed, Usman**; Roorda, Matthew 2021. Freight Vehicle Type Choice using Random Forest Classification. *Presented at the 56th Annual Conference of Canadian Transportation Research Forum, May 17-18, 2021 (Virtual Conference).*
- A1 Xu, Junshi; Tu, Ran; **Ahmed, Usman**; Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew 2021. How to Fairly Evaluate Eco-Driving? An Eco-Score System

Developed Using Vehicle Characteristics and Traffic Conditions. *Presented at the 56th Annual Conference of Canadian Transportation Research Forum, May 17-18, 2021.*

RESEARCH GRANTS

Year	Funding Organization	Team	Title	Amount (Total)
2025	Center for Regional and Rural Connected Communities	PI: Asad Khattak Co-PIs: Usman Ahmed , Jerry Everett, Venkatesh Pandey, Sudhagar Nagarajan, Xu Zhang	Techniques for Assessing Community Technology Readiness for Rural Development: Planning for Community Readiness Using Digital Twins.	\$656,911
2025	Center for Freight Transportation for Efficient & Resilient Supply Chain (UTK)	PI: Oriana Calderon, Co-PIs: Usman Ahmed , Katherine Asmussen	Spatial Modeling to support Supply Chain Policymaking across Metropolitan Areas in Tennessee.	\$31,186
2023	Region of Peel, Natural Science and Engineering Research Council of Canada	PI: Matthew Roorda Co-Investigator: Usman Ahmed	Smart Freight Centre: Freight Data Warehouse - Shipper Based Commercial Travel Survey.	\$120,000 (CAD)
2022	Munich Data Science Institute	PI: Rolf Moeckel Co-PI: Maximilian Schiffer External Collaborator: Usman Ahmed	Explainable AI to understand the impact of life events on travel behavior change (XAITraB)	€47,622

MENTORING AND SUPERVISING ACTIVITIES

Current Students

- Momina Liaqat Ali, PhD Student, Industrial and Systems Engineering, UTK
- Galen Flanagan, PhD Student, Data Science & Engineering, Bredesen Center, UTK
- Sharika Kabir, PhD Student, Civil Engineering, UTK
- Pranav Govindu, Undergraduate Researcher, Computer Science, UTK

Former Students

- Farah Ghizzawi, PhD candidate, Civil Engineering, University of Toronto
- Lisa Guseva, Undergraduate Researcher, Civil Engineering, University of Toronto

INVITED PRESENTATIONS

- **Ahmed, Usman.** Freight Surveys and Modeling - Lessons from Toronto. Invited Presentation to the Tennessee Model Users Group Meeting (TNMUG), Mt. Juliet, TN, May 13, 2025.
- **Ahmed, Usman.** Activity-Based Travel Demand Modeling in Toronto. Invited Presentation to CE558 - Transportation Planning Course, Department of Civil & Environmental Engineering, University of Tennessee Knoxville, April 23, 2025.

- **Ahmed, Usman.** Electric Cargo Pilot Study in Toronto. ITE Toronto Project of the Year Award, Feb 21, 2024, University of Toronto
- **Ahmed, Usman.** Transforming Last-Mile Delivery: Sustainable and Efficient Innovation. Invited panel discussion at the Fast Track Cities: Building Sustainable Supply Chains Event organized by CityAge, Feb 15, 2024.
- **Ahmed, Usman.** Greater Toronto and Hamilton Area - Commercial Travel Survey. Invited presentation to the research-in-progress meeting, Centre for Automated and Transformative Transportation Systems, University of Toronto, Jan 26, 2023.
- **Ahmed, Usman.** Establishment Location Choice Model Considering Intra-Firm Interactions. Invited presentation to the Integrated Land-use and Transportation Environment (ILUTE) Research Group at the University of Toronto, Nov 9, 2022.
- Xu, Junshi; Tu, Ran; **Ahmed, Usman**; Amirjamshidi, Glareh; Hatzopoulou, Marianne; Roorda, Matthew. An Eco-Score System Incorporating Driving Behavior, Vehicle Characteristics, and Traffic Conditions. Invited presentation to Mobility Chair Meeting, Ecole Polytechnique de Montreal, May 25-26, 2021.
- **Ahmed, Usman.** Urban Freight Transportation Model for GTHA, Invited Guest lecture to CIV1506: Freight Transportation and ITS Applications, March 22, 2021.
- **Ahmed, Usman.** Modelling Freight Vehicle Type using Machine Learning and Discrete Choice Methods, Invited Presentation to University of Toronto Student Chapter of ITE, March 12, 2021.

RESEARCH INTERESTS

- Agent-based modeling
- Supply chain and logistics
- Urban goods movement
- Travel behavior modeling
- Emerging transportation technologies
- Artificial intelligence, machine learning, big data

SKILLS

- **Methodological:** Discrete choice modeling, machine learning, optimization, spatial analysis, Bayesian statistics
- **Programming languages:** R, Python, MATLAB
- **GIS:** ArcGIS & QGIS
- **Transportation modeling:** EMME, VISUM, VISSIM
- **Transportation geometric design:** AutoCAD & Autodesk CIVIL 3D

SERVICE

Journal Referee Work: Transportation Research Record; Transportation Research Part A, B, C & E; Journal of Transport & Land Use; International Journal of Transportation Science & Technology; Transportation Letters: The International Journal of Transportation Research; Journal of Transport Geography; Research in Transportation Business & Management.

Conference Referee Work: Transportation Research Board (Since 2021); Models & Technologies for Intelligent Transportation Systems Conference (2023); Annual Bridging Transportation Researchers Conference (2022 & 2023); Canadian Society for Civil Engineering (2024).

Graduate Applications Review: Review of scholarship applications for Bredesen Center for Interdisciplinary Research and Graduate Education at the University of Tennessee, Knoxville.

Conference & Event Leadership

- **Session Chair of Travel Behavior, Equity, and Sustainability I session**, 16th World Conference on Transport Research, July 17-21, 2023, Montreal, Canada.
- **Technical Tour Lead**, 58th Annual CTRF Conference May 7-10, 2023, Hyatt Regency, Toronto, ON
- **Session Moderator**, 9th International Urban Freight Conference May 25-27, 2022, Long Beach, California.
- **Moderator**, Research in Progress Meeting, 2022 CLUE, University of Toronto
- **Moderator**, UT-ITE Weekly Webinar Series, 2021-2022, University of Toronto

Professional Memberships

- **Member**, American Society of Civil Engineers (ASCE)
- **Member**, Standing Committee on Freight Data (AED13), Transportation Research Board. August 16th, 2025 - April 14th, 2028.
- **Member**, Standing Committee on Freight Transportation Data (AED70), Transportation Research Board. April 15th, 2024 - May 30th, 2025.
- **Member**, Standing Committee on Statewide/National Transportation Data & Information Management (AED10), Transportation Research Board. April 15th, 2024 - May 30th, 2025
- **Member**, Subcommittee on Data, Planning, and Analysis Group Young Members (AE000(1)), Transportation Research Board. May 21st, 2024 - May 30th, 2025
- **Member**, Canadian Transportation Research Forum (2021-2022, 2024-2025)
- **Member**, Network on European Communications and Transport Activities Research (NECTAR) (2023)
- **President**, Institute of Transportation Engineers Student Chapter, University of Toronto (2021-2022)

- **Committee Member**, University of Toronto Transportation Alumni Network
- **Registered Engineer**, Pakistan Engineering Council (2014-2015)
- **Finance Officer**, Community Services Club-Art for Charity Chapter, National University of Sciences & Technology (2014)

PROFESSIONAL DEVELOPMENT

Workshops & Training Certifications

- T3. Regional I-Corps Program. University of Tennessee, Knoxville, Spring 2026
- T2. Planning More Equitable Transportation Systems, Mobility Network Summer School, University of Toronto, July 18-19, 2022.
- T1. ITS System Architecture Issues, Technical University of Munich, January 25-29, 2016.